From Molecule to Megabyte

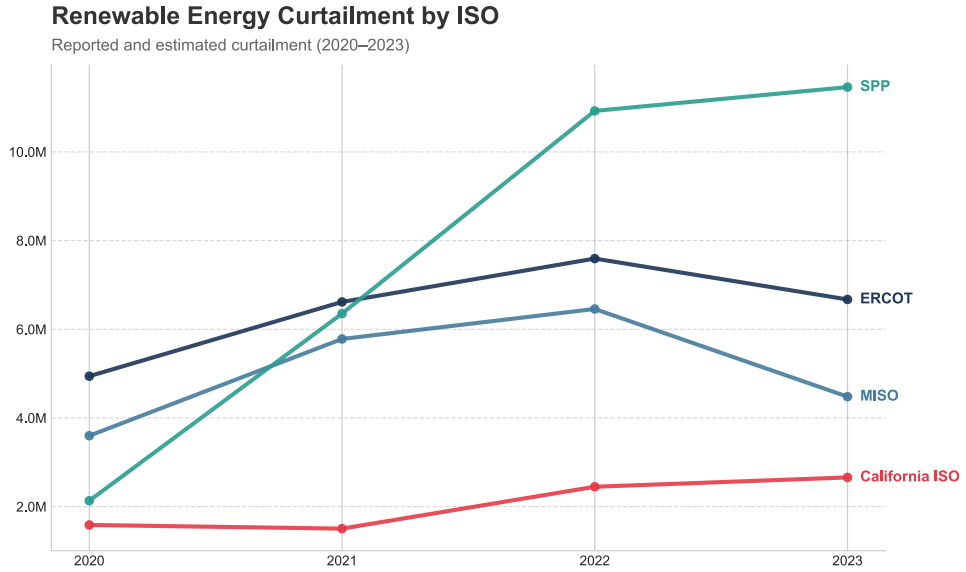
Sierra Solutions

SPE E3S Energython

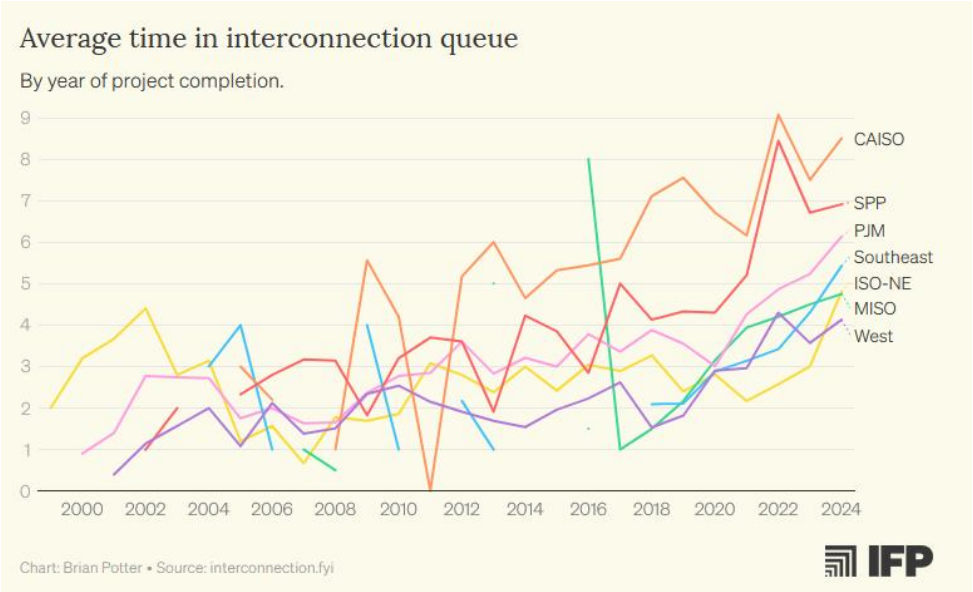
AI is Growing. Grid is *Not*.

The physical bottleneck of the digital revolution.

- AI demand is doubling every six months.
- Interconnection queues for grid power now exceed **60 months**.
- Curtailments increased **from 40% up to 500%** in different ISOs.
- **Grid curtailments** & **transmission backlogs** are a **death sentence for high-uptime AI inference**.



spglobal.com



Interconnection.fyi

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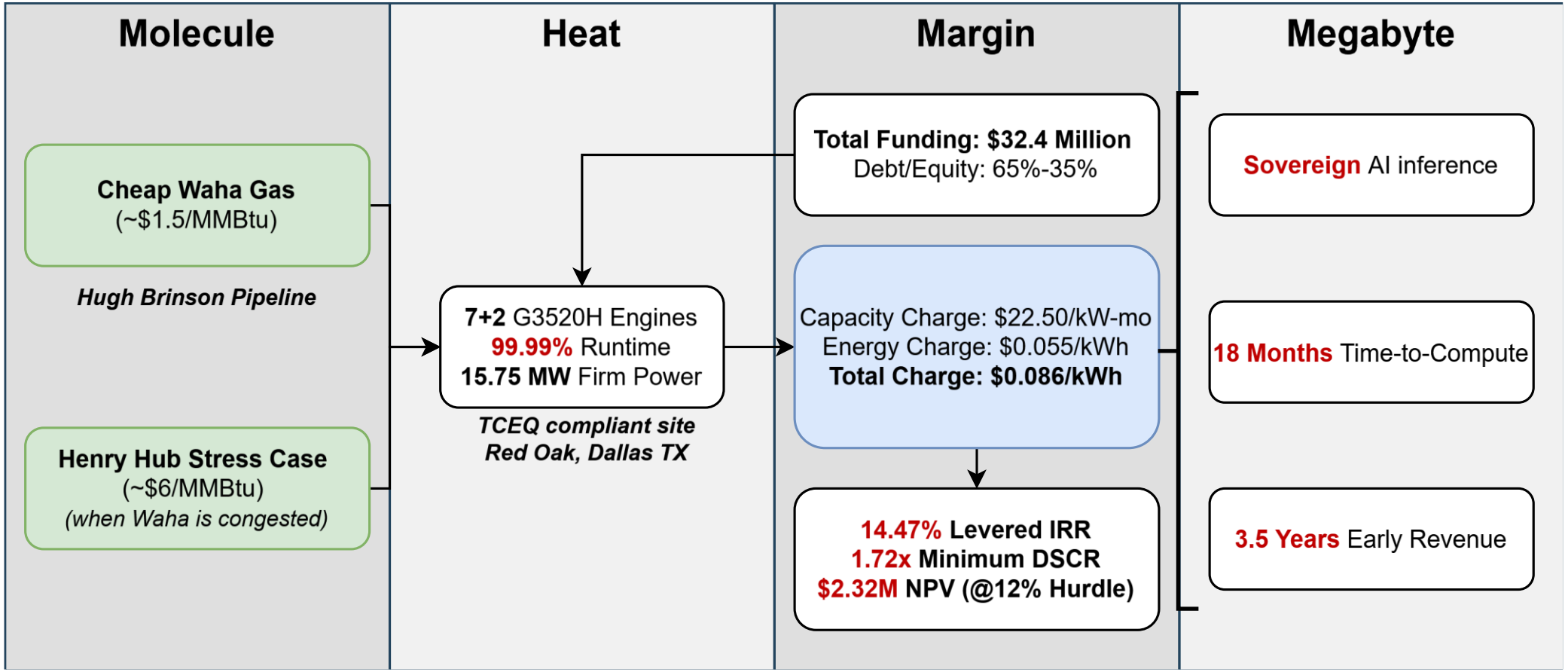
From Molecule to Megabyte

A high-reliability on-site power solution by Sierra Solutions.



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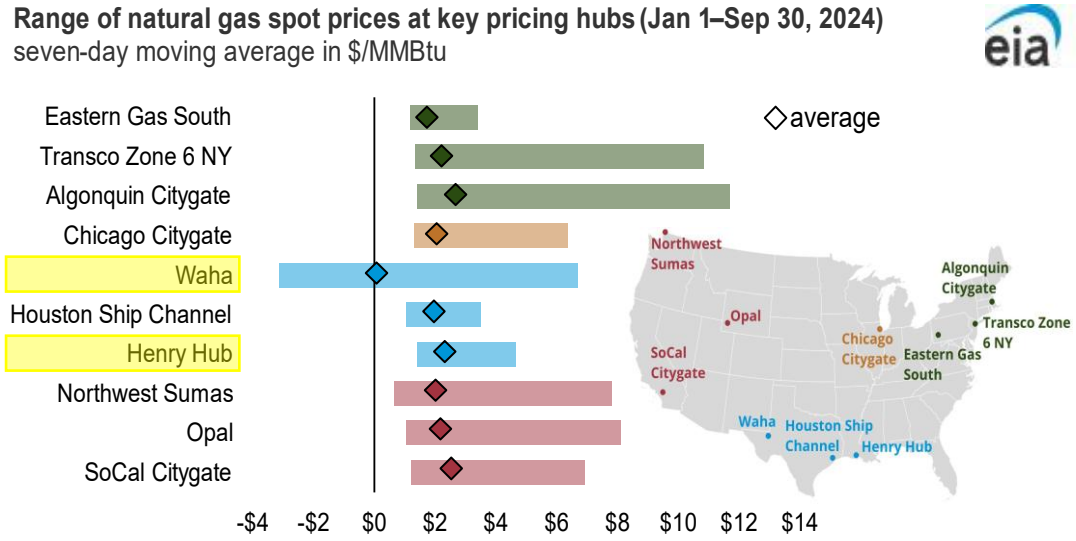
Why Dallas

High demand meeting physical gas optionality.

- Primary: Waha Hub (Cheapest in the US).
- Safety Net: Henry Hub (Most liquid in the US).
- Double-hub connectivity **eliminates fuel failure**.



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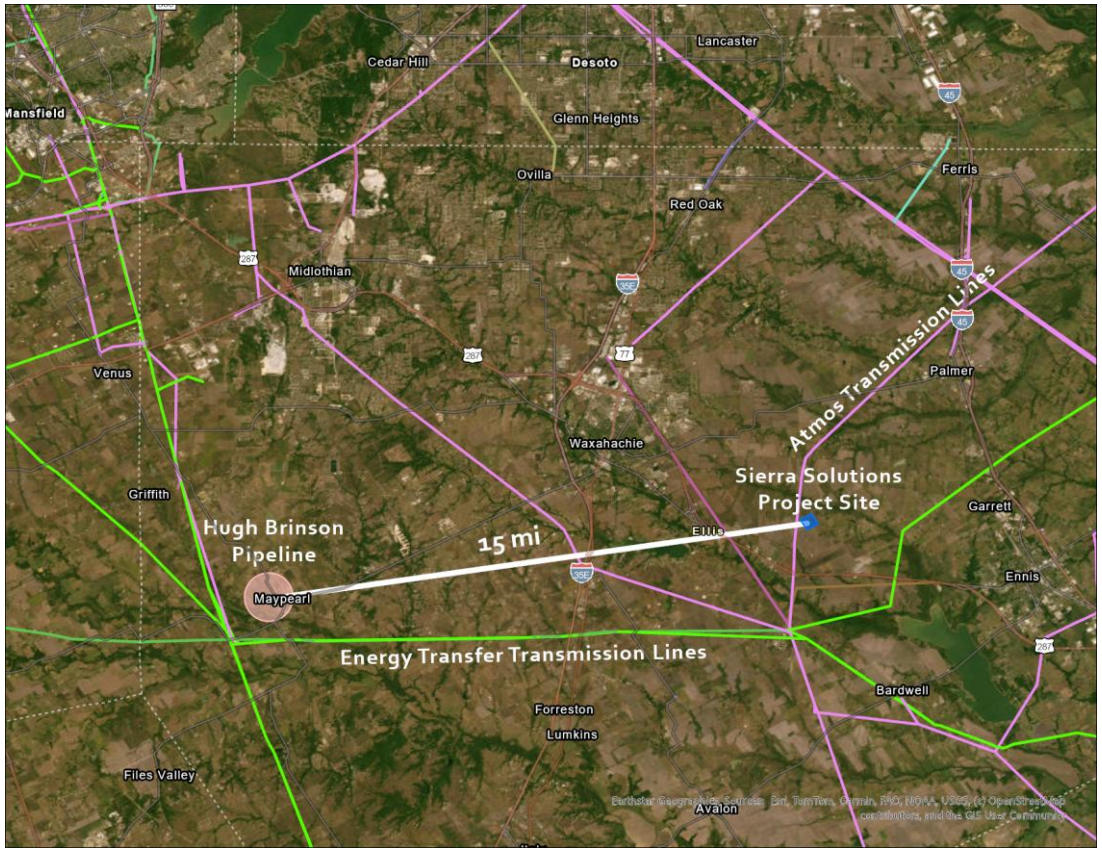

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Technology Refinement

Modular engines outperforming turbines for high-reliability 15MW loads.

		N Configuration		N+1 Configuration		N+2 Configuration		2N Configuration	
Name	Price ($M/unit)	Units (N)	Total CapEx ($M)	Units (N+1)	Total CapEx ($M)	Units (N+2)	Total CapEx ($M)	Units (2N)	Total CapEx ($M)
Titan 130	10.925	2	21.85	3	**32.78**	4	43.70	4	43.70
W31SG	17.655	2	35.31	3	**52.97**	4	70.62	4	70.62
G3520H	1.725	7	12.08	8	13.80	**9**	**15.53**	14	24.15

Name	Net Efficiency	Fuel Burn (MMBtu/hr)	Δ Fuel (MMBtu/hr)	Annual Fuel Penalty ($M/yr)	Project Fuel Penalty ($M)	Power, Net (MW)
Titan 130	0.354	144.58	45.39	0.60	8.95	11.92
W31SG	0.516	99.19	0.00	0.00	0.00	10.26
G3520H	0.421	121.57	22.38	0.29	4.41	2.25

westernstatescat.com

- 9 modular CAT G3520H units.
- N+2 Configuration: 15.75 MW Firm / 20.25 MW Installed.
- **Physics-based modeling:** 8,144 Btu/kWh heat rate with 0.5% annual mechanical degradation.

Project CAPEX

Detailed breakdown of hard and soft infrastructure costs indicates industrial average.

	Item	Qty	Unit	Unit Rate ($)	Total Cost ($)	Source / Logic
Power Generation Modules	CAT G3520H Generator Set (Skid)	9	Each	$1,725,000	$15,525,000	Market Quote (Surplus Record, 2025 Avg).
	Sound Attenuated Enclosure (ISO)	9	Each	$175,000	$1,575,000	MachineryTrader listing for 40' container w/ 85dBA rating.
	SCR Emissions Package	9	Each	$225,000	$2,025,000	EPA Retrofit Data ($100/kW) for NOx compliance.
Electrical BoP	15kV Switchgear Sections	12	Cell	$75,000	$900,000	Siemens GM-SG Style Vacuum Breaker (Installed New).
	Master Control Panel (Sync)	1	Lot	$650,000	$650,000	Complex Logic for 9-Unit Paralleling (Est. based on CAT EMCP 4.4).
	MV Cabling & Trays	3000	Ft	$150	$450,000	15kV Shielded Cable + Tray Install ($150/ft installed).
	Step-Up Transformers (2.5 MVA)	5	Each	$120,000	$600,000	Padmount, 13.8kV/480V. One per 2 engines + spare.
	Grid Connection Hardware	1	Lot	$400,000	$400,000	Overhead line + disconnect switch to Data Center
Mechanical BoP	Gas Fuel Train (Regulators)	9	Each	$35,000	$315,000	High-pressure regulators per engine inlet.
	Lube Oil Storage System	1	Lot	$150,000	$150,000	Central 5,000 gal tank + piping loop.
	Gas Interconnection Lateral	1	Mile	$1,000,000	$1,000,000	6" Lateral to Atmos/ET Trunk Line.
Civil & Site Works	Concrete Equipment Pads	9	Each	$25,000	$225,000	15 cy reinforced concrete per pad @ $1,500/cy.
	Site Grading & Fencing	4	Acre	$75,000	$300,000	Heavy grading + Anti-climb security fence.
	Land Acquisition	4	Acre	$250,000	$1,000,000	Red Oak Industrial Park Listing (2025).
	Spare Parts	1	Lot	$250,000	$250,000	Critical spares for Day 1 reliability.
Soft Costs	Engineering & Permitting	1	Lot	$1,500,000	$1,500,000	Air Permit (TCEQ) + Elect. Interconnect Study.
	Construction Management	1	Lot	5%	$1,343,250	5% of Hard Costs.
	Contingency (P50)	1	%	10%	$2,820,825	Standard Project Finance Reserve.
	TOTAL PROJECT COST				**$31,029,075**	
	$/kW (Installed)				**$1,532**	

- Includes $2.8M in P50 Contingency and $1.5M in Permitting/Engineering.
- $1532/kW is within the industry-reported range of $900-$2600/kW.[1]

[1] nrel.gov

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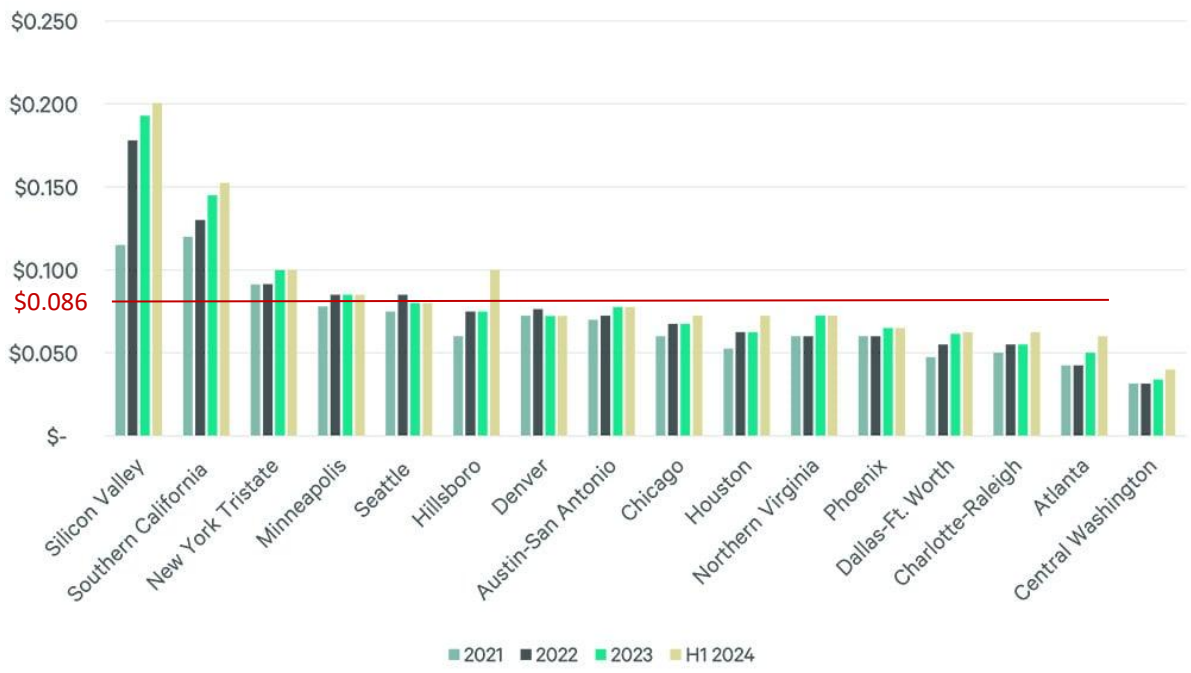
Reliability is Affordable



cbre.com

A Two-Part Tariff designed to protect debt service and equity upside.

- Capacity Rate: **$22.50 / kW-mo.**
  - (Reservation fee for 99.99% uptime).
- Energy Rate: **$0.055 / kWh.**
- 8.6 cents/kWh blended rate—highly competitive for on-site prime power.
  - 20%+ cheaper than high-cost state grids.
  - 1.3 cents/kWh higher than Dallas avg.
  - Avg includes grid rates where 99.99% uptime not guaranteed.

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Financial Rigor

Accounting for real-world applications with attention to detail.

- Precise escalation mapping for labor (3%), fuel (3%) and commodities (2.5%). Optimized MACRS 5-Year schedule starting at "Placed in Service" date.

Year	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Date	1-Jan-27	1-Jan-28	1-Jan-29	1-Jan-30	1-Jan-31	1-Jan-32	1-Jan-33	1-Jan-34	1-Jan-35	1-Jan-36	1-Jan-37	1-Jan-38	1-Jan-39	1-Jan-40	1-Jan-41	1-Jan-42	1-Jan-43	1-Jan-44
Energy Rev.	$0.00 M	$0.00 M	$3.85 M	$7.84 M	$8.08 M	$8.32 M	$8.57 M	$8.83 M	$9.09 M	$9.37 M	$9.65 M	$9.94 M	$10.23 M	$10.54 M	$10.86 M	$11.18 M	$11.52 M	$11.86 M
Capacity Rev.	$0.00 M	$0.00 M	$2.12 M	$4.31 M	$4.42 M	$4.53 M	$4.64 M	$4.76 M	$4.87 M	$5.00 M	$5.12 M	$5.25 M	$5.38 M	$5.51 M	$5.65 M	$5.79 M	$5.94 M	$6.09 M
Total Rev.	$0.00 M	$0.00 M	$5.98 M	$12.15 M	$12.49 M	$12.85 M	$13.21 M	$13.58 M	$13.97 M	$14.36 M	$14.77 M	$15.18 M	$15.61 M	$16.06 M	$16.51 M	$16.98 M	$17.46 M	$17.95 M
Total Fuel Cost	$0.00 M	$0.00 M	$1.32 M	$2.67 M	$2.75 M	$2.83 M	$2.90 M	$2.99 M	$3.07 M	$3.16 M	$3.25 M	$3.34 M	$3.44 M	$3.54 M	$3.65 M	$3.75 M	$3.86 M	$3.98 M
Variable O&M	$0.00 M	$0.00 M	$1.22 M	$2.47 M	$2.53 M	$2.59 M	$2.66 M	$2.72 M	$2.79 M	$2.86 M	$2.93 M	$3.00 M	$3.08 M	$3.16 M	$3.23 M	$3.32 M	$3.40 M	$3.48 M
Staffing	$0.00 M	$0.00 M	$0.49 M	$0.99 M	$1.02 M	$1.05 M	$1.09 M	$1.12 M	$1.15 M	$1.19 M	$1.22 M	$1.26 M	$1.30 M	$1.34 M	$1.38 M	$1.42 M	$1.46 M	$1.50 M
Ins. & Admin	$0.00 M	$0.31 M	$0.32 M	$0.33 M	$0.34 M	$0.34 M	$0.35 M	$0.36 M	$0.37 M	$0.38 M	$0.39 M	$0.40 M	$0.41 M	$0.42 M	$0.43 M	$0.44 M	$0.45 M	$0.46 M
Property Tax	$0.00 M	$0.78 M	$0.78 M	$0.74 M	$0.70 M	$0.67 M	$0.63 M	$0.60 M	$0.57 M	$0.54 M	$0.51 M	$0.49 M	$0.46 M	$0.44 M	$0.42 M	$0.40 M	$0.38 M	$0.36 M
Total Opex	$0.00 M	$1.08 M	$4.12 M	$7.20 M	$7.33 M	$7.48 M	$7.63 M	$7.79 M	$7.95 M	$8.13 M	$8.31 M	$8.49 M	$8.69 M	$8.89 M	$9.10 M	$9.32 M	$9.55 M	$9.79 M
Beginning Bal.	$0.00 M	$21.05 M	$22.84 M	$23.80 M	$22.96 M	$22.05 M	$21.05 M	$19.98 M	$18.81 M	$17.54 M	$16.17 M	$14.67 M	$13.05 M	$11.30 M	$9.39 M	$7.32 M	$5.08 M	$2.64 M
Interest	$0.00 M	$1.79 M	$1.94 M	$2.02 M	$1.95 M	$1.87 M	$1.79 M	$1.70 M	$1.60 M	$1.49 M	$1.37 M	$1.25 M	$1.11 M	$0.96 M	$0.80 M	$0.62 M	$0.43 M	$0.22 M
Cap Int (IDC)	$0.00 M	$1.79 M	$0.97 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M
Principal PMT	$0.00 M	$0.00 M	$0.00 M	$0.84 M	$0.91 M	$0.99 M	$1.08 M	$1.17 M	$1.27 M	$1.38 M	$1.49 M	$1.62 M	$1.76 M	$1.91 M	$2.07 M	$2.24 M	$2.44 M	$2.64 M
End Balance	$21.05 M	$22.84 M	$23.80 M	$22.96 M	$22.05 M	$21.05 M	$19.98 M	$18.81 M	$17.54 M	$16.17 M	$14.67 M	$13.05 M	$11.30 M	$9.39 M	$7.32 M	$5.08 M	$2.64 M	$0.00 M
EBITDA	$0.00 M	-$1.08 M	$1.86 M	$4.95 M	$5.16 M	$5.37 M	$5.58 M	$5.79 M	$6.01 M	$6.23 M	$6.46 M	$6.69 M	$6.92 M	$7.16 M	$7.41 M	$7.65 M	$7.91 M	$8.16 M
MACRS	$0.00 M	$0.00 M	$6.21 M	$9.93 M	$5.96 M	$3.57 M	$3.57 M	$1.79 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M
Taxable Income	$0.00 M	-$2.87 M	-$6.29 M	-$7.00 M	-$2.75 M	-$0.08 M	$0.22 M	$2.31 M	$4.41 M	$4.74 M	$5.09 M	$5.44 M	$5.81 M	$6.20 M	$6.61 M	$7.03 M	$7.47 M	$7.94 M
Income Tax	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.00 M	$0.05 M	$0.49 M	$0.93 M	$1.00 M	$1.07 M	$1.14 M	$1.22 M	$1.30 M	$1.39 M	$1.48 M	$1.57 M	$1.67 M
Debt Service	$0.00 M	$0.00 M	$0.98 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M	$2.87 M
LFCF	-$11.33 M	-$1.08 M	$0.89 M	$2.09 M	$2.29 M	$2.50 M	$2.67 M	$2.44 M	$2.22 M	$2.37 M	$2.53 M	$2.68 M	$2.84 M	$2.99 M	$3.15 M	$3.31 M	$3.47 M	$3.63 M
DSCR			1.91	1.72	1.80	1.87	1.93	1.85	1.77	1.83	1.88	1.94	1.99	2.04	2.10	2.15	2.21	2.27

MACRS Year	Rate (%)
1	20
2	32
3	19.2
4	11.52
5	11.52
6	5.76
+7	0

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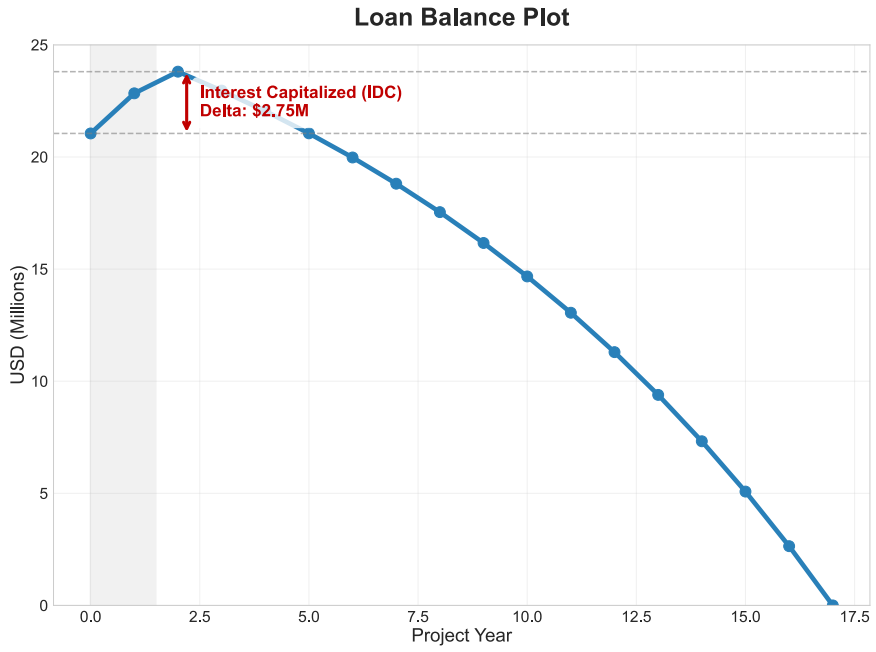
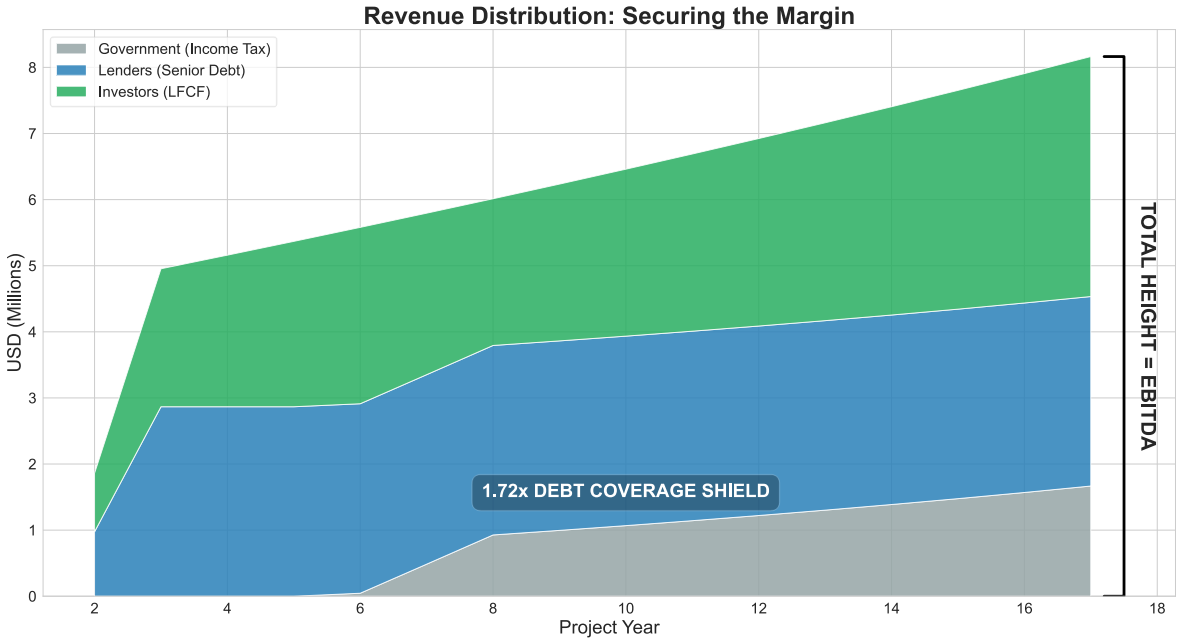
Bankable Margins

Visualizing the 1.72x DSCR shield.

- $2.75M in interest capitalized during the 18-month build.
- Capacity payments alone cover 100% of debt obligations.
- MACRS tax shield maintains $0 liability until year 6.





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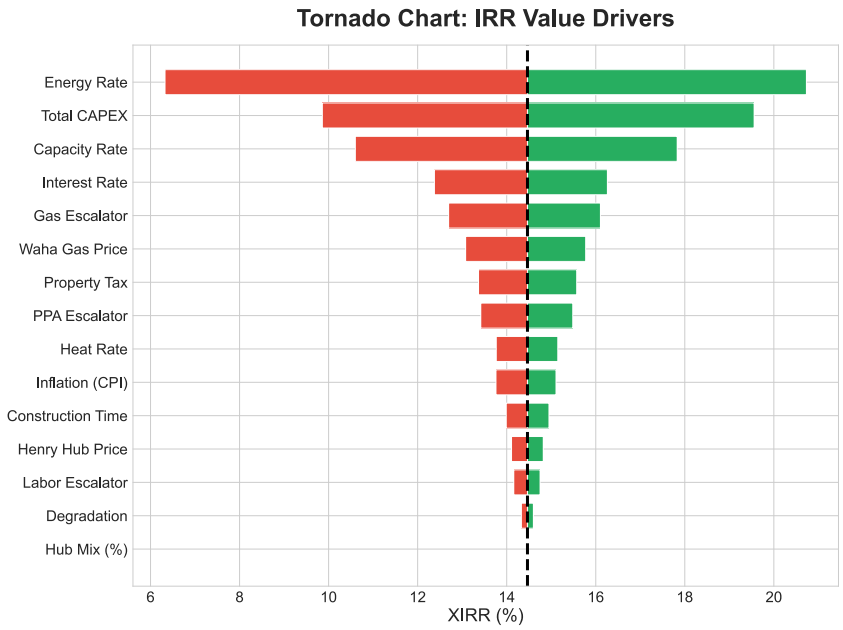
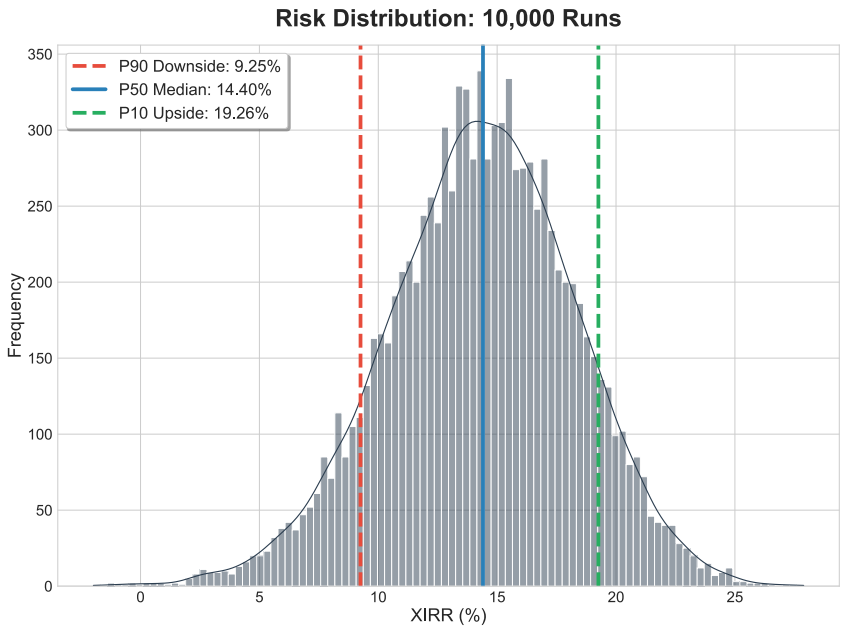
Value Driver Analysis

Stress-testing the 14.47% IRR against 12 operational levers.

- Primary value is driven by the **commercial spread** (Energy Rate).
- P90 of 9.3% IRR—Equity protected in "Perfect Storm" scenarios.
- P10 potential of 19.3% if Waha gas oversupply persists (Very Likely).





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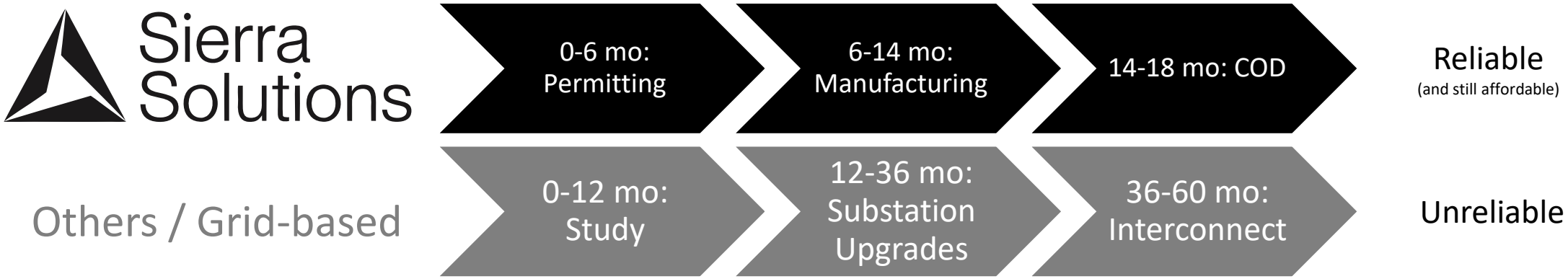
Winning the Race to Data

Speed-to-market as the ultimate competitive advantage.

- Sierra Solutions bypasses the rising interconnection queue (4+ years in ERCOT, 5+ years average).
- Our clients gain 30-42 months of extra server revenue vs. grid-connected competitors.
- Providing "Sovereign Power" for AI inference workloads that cannot afford grid volatility.



The Ask

Seeking $11.3M in equity to break ground on Project Dallas.

- 14.47% Levered IRR | $2.32M NPV | 1.72x DSCR.
- Breaking the utility bottleneck for the DFW high-tech sector.
- Join Sierra Solutions in **bridging the gap from Molecule to Megabyte**.

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Thank you

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Q&A Slides

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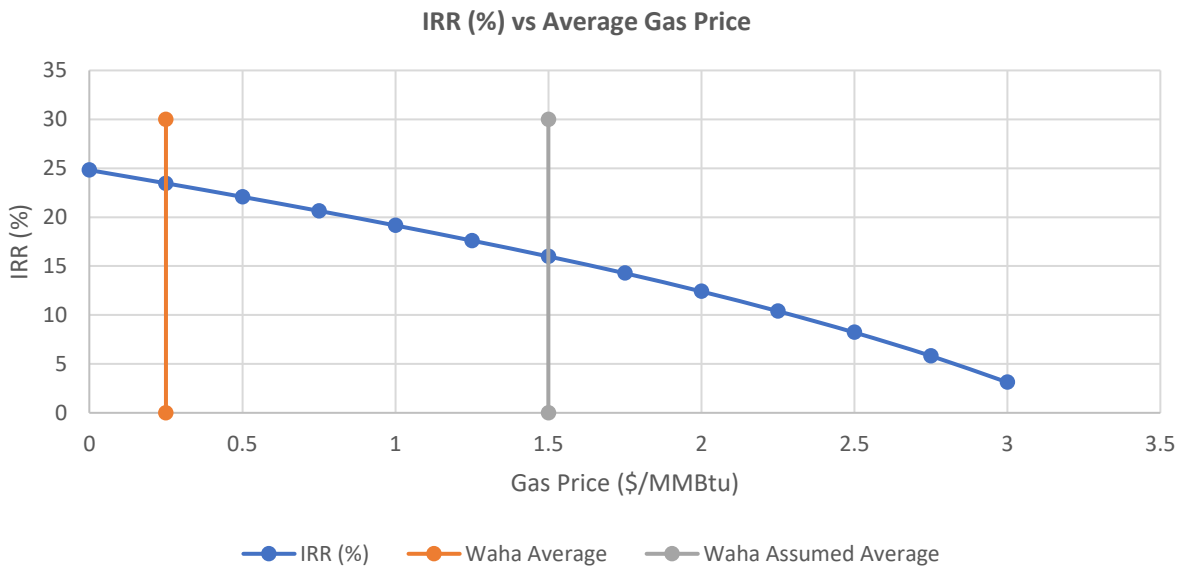
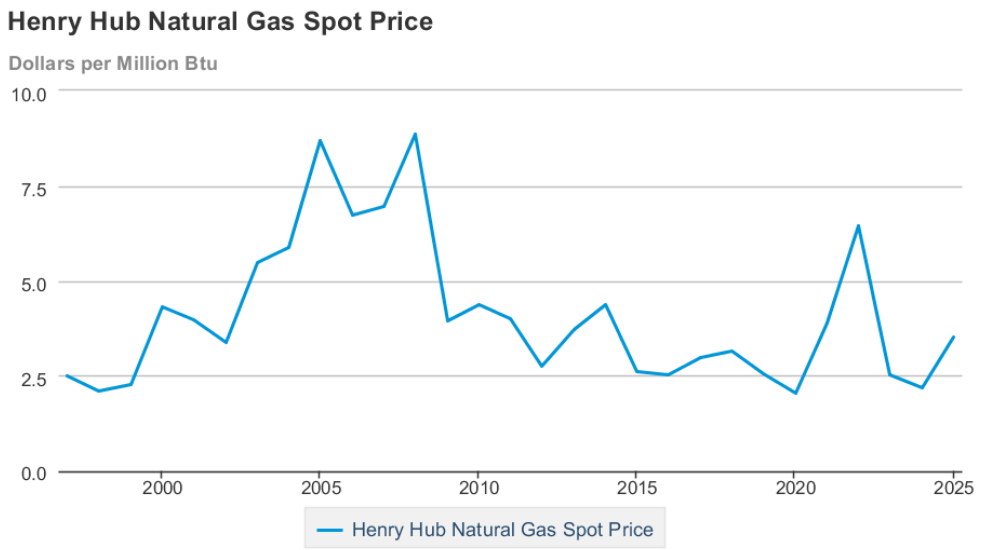
SPE E3S Energython

Strategic Resilience: Fuel Conservatism

Impact of regional price parity on project IRR.

- Our 14.5% IRR is based on $1.5/MMBtu gas price – **6 times higher** than Waha average.
- At historical average of $0.25/MMBtu, Project Dallas delivers a **23% IRR**.
- **9% IRR** if prices between Waha-Henry comes to an equilibrium at $2.5/MMBtu.





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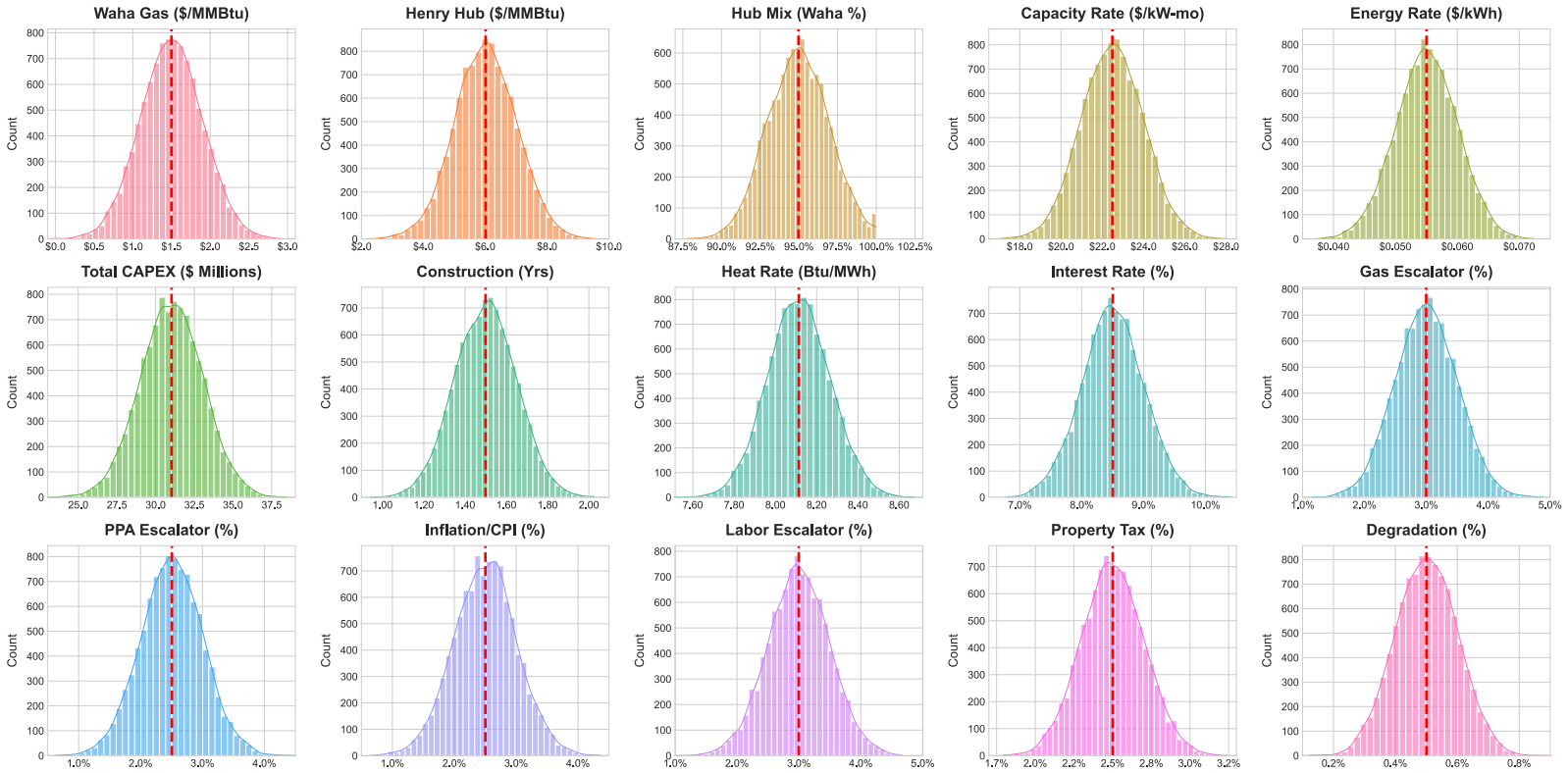
Stochastic Boundary Conditions

Defining the Stress Portfolio for 10000 Simulations

- Waha Gas: Subjected to a $**0.40 deviation (27% of mean)** to account for Texas regional pipeline constraints & seasonal price spikes.
- Henry Hub: Modeled with a **$1.00 volatility window** to reflect national liquidity shifts.
- Hard CAPEX: Stress-tested with a **$2.0M variance** to capture supply chain and labor inflationary pressures.
- Construction Timing: Modeled across **±5 month window**.
- Hub Mix: Varied from **90% to 100% Waha-sourcing window**.
- Interest Rates: **±50 basis points** to ensure debt serviceability.
- Escalators: All growth rates subjected to **±0.5% variance**.

Appendix: Project Stress-Test Assumptions (Full 15-Variable Mapping)



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Data Sources: Financial & Commercial Inputs

Benchmarking cost-basis and revenue assumptions against 2025 market data.

- CAT G3520H Genset Price ($1,725,000): Sourced from Surplus Record (2025 Avg) market quotes.
- Sound Enclosure ($175,000): Sourced from MachineryTrader listings for 85dBA rated ISO containers
- SCR Emissions Package ($225,000): Sourced from EPA Retrofit Data (approx. $100/kW for NOx compliance).
- 15kV Switchgear ($75,000/cell): Pricing based on Siemens GM-SG Style Vacuum Breakers.
- Land Acquisition ($250,000/acre): Sourced from Red Oak Industrial Park real estate listings (2025).
- MV Cabling ($150/ft): Industry estimate for 15kV shielded cable + tray installation.
- Concrete/Civil Works: Unit rates ($1,500/cy) based on standard civil engineering estimates.
- Electricity Price Benchmarks ($0.06 - $0.09/kWh) sourced from CBRE Data Center Trends reports.
- Hurdle Rate (12%) and DSCR Target (1.25x) sourced from SPE Competition Briefing (Stephen Janssen).
- Tax Depreciation (MACRS 5-Year) based on IRS Publication 946 for eligible power generation property.

Data Sources: Technical & Operational Inputs

Physics-based modeling parameters anchored in OEM specifications and federal data.

- Engine Performance (8,114 Btu/kWh LHV) sourced directly from Caterpillar G3520H Spec Sheets.
- Interconnection Queue delays (60+ Months) sourced from Interconnection.fyi and LBNL Empiricist Reports.
- Grid Curtailment trends sourced from S&P Global Market Intelligence and EIA.gov.
- Natural Gas Spot Prices (Waha and Henry Hub) sourced from EIA Historical Data Series.
- Net-to-ISO Derating Factor (0.90) applied to account for Dallas ambient temperatures (105°F) and parasitic loads.
- Annual Mechanical Degradation (0.5%) based on standard reciprocating engine lifecycle curves.
- Firm Transport Fees ($0.60/MMBtu) based on standard intrastate Texas pipeline tariffs.
- Property Tax Abatement structure modeled on Dallas County Chapter 312 economic incentives.

Competitive Analysis: Technology Selection Data

Comparative CAPEX analysis driving the decision for Modular Reciprocating Engines.

- Solar Titan 130 Turbine pricing (10.925M per unit) sourced from distributed generation market reports.
- Wartsila 31SG pricing (17.655M perunit) derived from techno-economic literature on flexible gas power plants.
- Operational Flexibility: CAT G3520H selected for superior start-up time (minutes) vs. Frame Turbines (hours) to meet AI inference load volatility.

Engine Room

Validating investment integrity through a synced Python-Excel framework.

- **Excel Architecture:** Serving as the static "Ground Truth" for all engineering, commercial, and financial inputs to ensure auditability.
- **Python Integration:** Custom "Digital Twin" script reads Excel inputs dynamically to perform 15-year cash flow waterfall calculations.
- **Stochastic Layer:** Python engine executes 10,000 Monte Carlo iterations using the Excel base case as the statistical mean.

Project Inputs		Finance Model								MACRS Table	
Model Start Date	1-Jan-27	**Investment**		**Fuel Supply**		**Revenue & OpEx**		**Power Generation**		**MACRS Year**	**MACRS Rate (%)**
Construction period (years)	1.5	Debt (fraction)	0.65	Gas Price Waha ($/MMBtu)	1.5	Capacity Rate ($/kW-mo)	$22.50	Active # of Turbines	7	1	20
Operation period (years)	15	Equity (fraction)	0.35	Gas Price Henry ($/MMBtu)	6	Energy Rate ($/kWh)	$0.055	Total # of Turbines	9	2	32
Delivered Power (MW)	15	Loan Interest Rate (%)	8.5	Gas Fraction Waha	0.95	Income Tax (%)	21	Unit Price ($)	$1,725,000.00	3	19.2
		DSRA ($)	$1,355,000	Gas Fraction Henry	0.05	Base Insurance (%)	0.75	Unit Net Power (MW)	2.25	4	11.52
		Hard Capex ($)	$31,029,075.00	Avg Gas Price ($/MMBtu)	1.725	Base Admin ($)	$75,000.00	Peak power (MW)	20.25	5	11.52
		Initial Loan	$21,049,648.75	Gas Transport Fee ($/MMBtu)	0.6	Variable O&M Base (%)	17.5	Size (acres)	4	6	5.76
		Property				Staffing Base ($)	$923,000.00	Capacity Factor	0.96	7	0
		Land Price (Acre/MW)	0.2			**Escalators**		Heat Rate (Btu/MWh)	8.114		
		Property Tax (%)	2.5			Inflation (CPI) (%)	2.5	HHV/LHV (%)	10.8		
		Tax Drop (%)	5			PPA Escalator (%)	2.5	Degradation Factor (%)	0.5		
						Wage Escalator (%)	3				
						Gas Escalator (%)	3				

```python
def run_dallas_twin_model(p):
    # Sourcing Strategy
    weighted_hub_price = (0.95 * p['waha_gas_base']) + (0.05 * p['henry_gas_base'])
    initial_loan_draw = (p['capex_hard_soft'] + p['dsra']) * p['debt_share']
    initial_equity_y0 = (p['capex_hard_soft'] + p['dsra']) * (1 - p['debt_share'])

    # MACRS 5-Year Table (20%, 32%, 19.2%, 11.52%, 11.52%, 5.76%)
    macrs_table = [0, 0.20, 0.32, 0.192, 0.1152, 0.1152, 0.0576] + [0]*40
    first_full_ops_year = int(np.ceil(p['const_years'])) + 1

    results = []
    curr_loan_bal = initial_loan_draw
    balloon_bal = 0

    for yr in range(p['total_range']):
        # YOUR VERIFIED LOGIC: Switching and Timing
        rf = max(0, min(1, yr - p['const_years'])) if yr > 0 else 0
        ti = (max(yr-1, p['const_years']) + yr)/2 if rf > 0 else 0

        # Revenue
        total_rev = ((p['gen_mwh'] * 1000 * p['en_rate_base'] * ((1 +
p['gas_esc'])**ti)) + \
                     (p['mw_delivered'] * 1000 * p['cap_rate_base'] * 12 * ((1 +
p['ppa_esc'])**ti))) * rf

        # Opex
        fuel = p['gen_mwh'] * (p['hr'] * ((1 + p['degrad_rate'])**ti)) * \
               ((weighted_hub_price * ((1 + p['gas_esc'])**ti)) +
p['transport_fee']) * rf
        var_om = (p['gen_mwh'] * 17.50 * ((1 + p['cpi_esc'])**ti)) * rf
        staff = (p['staff_base'] * ((1 + p['labor_esc'])**ti)) * rf

        stat_on = 1 if yr > 0 else 0
        ins_adm = (p['ins_base'] + p['adm_base']) * ((1 + p['cpi_esc'])**ti) if rf >
0 else (p['ins_base'] + p['adm_base']) * stat_on
        tax_p = (p['capex_hard_soft'] * p['prop_tax']) * ((1 -
p['tax_drop'])**max(0, yr - 2)) * stat_on
        ebitda = total_rev - (fuel + var_om + staff + ins_adm + tax_p)

        # Debt Mechanics (Real-Shielded)
        interest = (curr_loan_bal * p['interest_rate']).real if yr > 0 else 0
        idc = interest * (1 - rf) if yr > 0 else 0
```

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